

Degeneracy Alone Does Not Force a Square-Root Gap in Finite-Horizon Restless Bandits

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Abstract

Gast, Gaujal, and Yan conjectured that, for essentially all degenerate finite-horizon restless bandits, every LP-compatible policy should have LP-relaxation gap of order $N^{-1/2}$. We give a small finite-horizon counterexample to the literal lower-bound statement. The instance has horizon $T = 2$, four states, a unique optimal LP solution, and is degenerate in the sense that no optimal LP solution has a split state at every time. Nevertheless a Lipschitz time-dependent priority policy is LP-compatible and, under the randomized rounding scheme of Gast–Gaujal–Yan, has exact gap

$$V_{\text{rel}}(m(0), 2) - V_{\pi}^{(N)}(m(0), 2) = \frac{\{\alpha N\}(1 - \{\alpha N\})}{N} \leq \frac{1}{4N}.$$

The example has genuinely stochastic individual transitions, but the stochastic fluctuation in the normal direction to the degenerate boundary is zero. Thus degeneracy must be supplemented by a nonzero transverse-variance assumption if one wants a universal square-root lower bound.

1 Statement of the point

We use the notation of Gast, Gaujal, and Yan [1]. A finite-horizon restless bandit has a finite state space S , two actions $a \in \{0, 1\}$, transition matrices P^a , rewards R_s^a , horizon T , and an activation fraction $\alpha \in (0, 1)$. Its LP relaxation is

$$\begin{aligned} V_{\text{rel}}(m(0), T) &= \max_{y \geq 0} \sum_{t=0}^{T-1} \sum_{s,a} R_s^a y_{s,a}(t) \\ \text{s.t. } y_{s,0}(t+1) + y_{s,1}(t+1) &= \sum_{s',a} y_{s',a}(t) P_{s's}^a, \quad s \in S, \ 0 \leq t < T-1, \\ \sum_s y_{s,1}(t) &= \alpha, \quad 0 \leq t < T, \\ y_{s,0}(0) + y_{s,1}(0) &= m_s(0), \quad s \in S. \end{aligned}$$

For an optimal LP solution y^* , write

$$S^0(t) = \{s \in S : y_{s,0}^*(t) > 0 \text{ and } y_{s,1}^*(t) > 0\}.$$

The model is called non-degenerate if there exists an optimal LP solution for which $|S^0(t)| \geq 1$ for all t ; otherwise it is degenerate. A policy π is LP-compatible if, along the fluid trajectory of some optimal LP solution, it reproduces that optimal LP solution.

The conjectural lower bound discussed in [1, Section 4.3.2] says, informally, that for essentially all degenerate restless bandits and all LP-compatible policies π , there should be constants $0 < \underline{C} \leq \overline{C} < \infty$, independent of N , such that

$$\frac{\underline{C}}{\sqrt{N}} \leq |V_{\pi}^{(N)}(m(0), T) - V_{\text{rel}}(m(0), T)| \leq \frac{\overline{C}}{\sqrt{N}}.$$

The construction below shows that the lower bound is false without an additional assumption.

Theorem 1. *For every $\alpha \in (0, 1)$ and every $p \in (0, 1)$, there is a finite-horizon restless bandit with $T = 2$ and four states such that:*

- (i) *the LP relaxation has a unique optimal solution;*
- (ii) *the model is degenerate;*
- (iii) *there is an LP-compatible Lipschitz priority policy π ; and*
- (iv) *under the randomized rounding procedure of [1], for all $N \geq 1$,*

$$V_{\text{rel}}(m(0), 2) - V_{\pi}^{(N)}(m(0), 2) = \frac{\{\alpha N\}(1 - \{\alpha N\})}{N}.$$

Consequently,

$$\sqrt{N} (V_{\text{rel}}(m(0), 2) - V_{\pi}^{(N)}(m(0), 2)) \longrightarrow 0,$$

and no positive $N^{-1/2}$ lower bound can hold for this degenerate instance.

2 The restless bandit instance

Fix $\alpha \in (0, 1)$ and $p \in (0, 1)$. Let

$$S = \{0, H_1, H_2, L\}.$$

The initial condition is

$$m(0) = e_0 = (1, 0, 0, 0),$$

and the horizon is $T = 2$. Rewards are action-dependent but time-homogeneous:

$$R_s^0 = 0 \quad \text{for all } s, \quad R_0^1 = 0, \quad R_{H_1}^1 = R_{H_2}^1 = 1, \quad R_L^1 = 0.$$

Thus reward is earned only by activating a high state, H_1 or H_2 .

With rows and columns ordered as $(0, H_1, H_2, L)$, define

$$P^1 = \begin{pmatrix} 0 & p & 1-p & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad P^0 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

An active arm in state 0 jumps randomly to one of the two high states; a passive arm in state 0 jumps to the low state L . All other states are absorbing. The individual transition from state 0 under action 1 is stochastic because $p \in (0, 1)$.

3 The LP optimum and degeneracy

At time 0, all mass is in state 0. The LP budget constraint therefore forces

$$y_{0,1}^*(0) = \alpha, \quad y_{0,0}^*(0) = 1 - \alpha,$$

with all other time-0 variables equal to zero. Hence the time-1 fluid state is

$$m^*(1) = (0, p\alpha, (1-p)\alpha, 1 - \alpha).$$

At time 1, the total mass in high states is exactly α . The reward from activating either high state is 1, whereas the reward from activating state L or state 0 is 0. Hence every unit of high-state mass must be activated in any optimal LP solution. Since the high-state mass equals the entire budget, the unique time-1 decision is

$$y_{H_1,1}^*(1) = p\alpha, \quad y_{H_2,1}^*(1) = (1-p)\alpha, \quad y_{L,0}^*(1) = 1 - \alpha,$$

with all remaining time-1 variables equal to zero. The LP value is therefore

$$V_{\text{rel}}(m(0), 2) = \alpha.$$

This also proves uniqueness of the optimal LP solution.

The split sets are

$$S^0(0) = \{0\}, \quad S^0(1) = \emptyset.$$

Thus there is no optimal LP solution with a split state at every time. Since the LP optimum is unique, the instance is degenerate.

4 An LP-compatible Lipschitz policy

For any priority order $\sigma = (\sigma_1, \dots, \sigma_4)$, define the priority activation map by

$$A_{\sigma_i}(m) = \min \left\{ m_{\sigma_i}, \left(\alpha - \sum_{j < i} A_{\sigma_j}(m) \right)_+ \right\},$$

$$\pi_{\sigma_i,1}^\sigma(m) = A_{\sigma_i}(m), \quad \pi_{\sigma_i,0}^\sigma(m) = m_{\sigma_i} - A_{\sigma_i}(m).$$

This map is admissible. Since it is obtained from finitely many additions, maxima, and minima, it is Lipschitz.

Let the policy $\pi = (\pi_0, \pi_1)$ use the priority order

$$0 \succ H_1 \succ H_2 \succ L \quad \text{at time 0,}$$

and

$$H_1 \succ H_2 \succ L \succ 0 \quad \text{at time 1.}$$

At the initial fluid state $m^*(0) = e_0$, this policy activates exactly an α fraction of state-0 mass. At the time-1 fluid state $m^*(1)$, it activates all mass in H_1 and H_2 and none in L . Therefore

$$\pi_t(m^*(t)) = y^*(t), \quad t = 0, 1,$$

so π is LP-compatible.

5 Exact finite- N calculation

Let

$$q = \lfloor \alpha N \rfloor, \quad r = \{\alpha N\} = \alpha N - q.$$

At time 0, the policy prescribes activation mass α in state 0. Randomized rounding activates

$$K_N = \begin{cases} q + 1, & \text{with probability } r, \\ q, & \text{with probability } 1 - r. \end{cases}$$

Every active state-0 arm moves to one of H_1, H_2 , and every passive state-0 arm moves to L . Hence the total number of high-state arms at time 1 is exactly K_N . The random split between H_1 and H_2 is irrelevant for the reward because both high states have the same active reward.

At time 1, the policy activates high states before low states. If $K_N = q$, then the high-state mass is $q/N \leq \alpha$, so all high-state arms are activated and the time-1 reward is q/N . If $K_N = q + 1$, then the high-state mass exceeds α , so the policy prescribes total high-state activation mass α ; randomized rounding activates high-state arms in expectation αN , and the conditional expected reward is α . There is no time-0 reward. Thus

$$\begin{aligned} V_\pi^{(N)}(m(0), 2) &= (1 - r) \frac{q}{N} + r\alpha \\ &= (1 - r) \frac{q}{N} + r \frac{q + r}{N} \\ &= \frac{q + r^2}{N}. \end{aligned}$$

Since $V_{\text{rel}}(m(0), 2) = \alpha = (q + r)/N$, the gap is

$$V_{\text{rel}}(m(0), 2) - V_\pi^{(N)}(m(0), 2) = \frac{q + r}{N} - \frac{q + r^2}{N} = \frac{r(1 - r)}{N} \leq \frac{1}{4N}.$$

In particular the gap is zero whenever αN is an integer, and in all cases it is $O(N^{-1})$.

6 Why the square-root mechanism is absent

The degenerate boundary at time 1 is the hyperplane

$$m_{H_1} + m_{H_2} = \alpha.$$

Although the active transition from state 0 is stochastic between H_1 and H_2 , the indicator of membership in the high set $H = \{H_1, H_2\}$ is deterministic under each relevant action:

$$\mathbb{P}(X_{t+1} \in H \mid X_t = 0, a = 1) = 1, \quad \mathbb{P}(X_{t+1} \in H \mid X_t = 0, a = 0) = 0.$$

Therefore the normal coordinate $m_{H_1} + m_{H_2} - \alpha$ has only the $O(1/N)$ randomized-rounding error from the number of active arms at time 0, not a central-limit $O(N^{-1/2})$ fluctuation. The usual square-root loss at a kink is caused by a nonzero one-sided shadow-price kink together with nonzero Gaussian-scale fluctuation across the kink. This example has the kink but not the transverse fluctuation.

The conclusion is consequently limited but sharp: degeneracy alone does not imply an $\Omega(N^{-1/2})$ lower bound. A generic theorem of that form would need an additional nonzero transverse-variance condition, or an equivalent condition ruling out variance-degenerate boundary crossings such as the one above.

Remark 1 (Three-state deterministic version). *Setting $p = 1$ and removing the unused state H_2 gives an even smaller three-state deterministic example. The four-state version above keeps a genuinely stochastic individual transition while preserving zero variance in the boundary-normal direction.*

References

- [1] N. Gast, B. Gaujal, and C. Yan. Linear program-based policies for restless bandits: Necessary and sufficient conditions for (exponentially fast) asymptotic optimality. *Mathematics of Operations Research*, 49(4):2468–2491. Published online 2023. doi: [10.1287/moor.2022.0101](https://doi.org/10.1287/moor.2022.0101). Preprint: arXiv:2106.10067.